FTL FOR ONLINE LINEAR OPTIMIZATION

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- No matter how we choose the first action a_1^{FTL}, it will hold that FTL has a cumulative loss greater than (or equal) $T - 3/2$, while the best action in hindsight has a cumulative loss of $-1/2$.
- Thus, FTL's cumulative regret is at least $T - 1$, which is linearly growing in T.

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- Indeed, note that

$$a_{t+1}^{\mathrm{FTL}} = \arg\min_{a \in \mathcal{A}} \sum_{s=1}^{t} \ell(a, z_s) = \arg\min_{a \in [-1,1]} a \sum_{s=1}^{t} z_s = a \sum_{s=1}^{t} z_s$$

$$= \begin{cases} -1, & \text{if } \sum_{s=1}^{t} z_s > 0, \\ 1, & \text{if } \sum_{s=1}^{t} z_s < 0, \\ \text{arbitrary}, & \text{if } \sum_{s=1}^{t} z_s = 0. \end{cases}$$

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t	a_t^{FTL}	z_t	$(a_t^{\mathrm{FTL}}, z_t)$	$\sum_{s=1}^{t} (a_s^{\mathrm{FTL}}, z_s)$	$\sum_{s=1}^{t} z_s$
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2	1	1	1	$1 - 1/2$	$1/2$

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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
T	$(-1)^T$	$(-1)^T$	1	$T - 1 - 1/2$	$(-1/2)^T$

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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
T	$(-1)^T$	$(-1)^T$	1	$T - 1 - 1/2$	$((-1/2)^T$

- The best action has cumulative loss

$$\inf_{a \in \mathcal{A}} \sum_{s=1}^{T} \ell(a, z_s) = \inf_{a \in [-1,1]} a \underbrace{\sum_{s=1}^{T} z_s}_{=((-1/2)^T} = -1/2.$$